

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	Slot	B
Course Title	Cloud Computing and Big Data Analytics	Assessment	CO4 AT1
Assessment Title	Big Data Analytics Class Test	Bloom Level	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	Total Marks	30

Student Details

Student Name	L Nihás Goud	Register Number	192472346
Class / Section	CSE-AI	Slot	B
Capstone Title	SkillHire — Cloud Resume-to-Job Matcher Using NLP Skill Gap Intelligence	Submission Date	19-08-2026
GitHub Repository Link	https://github.com/nihásgoud-stack/CSA1512_Assessment	Google Classroom Status	Submitted

Assessment Overview

CO4 Focus

Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.
- Use diagrams or small tables wherever they improve clarity.
- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Criteria	Marks	Evaluation Focus	Student Check
Concept Accuracy	9	Correct technical meaning, terminology and examples.	Have I used correct big data terms?
Coverage	6	All required parts of the question are answered.	Have I answered every part?

Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I **L. NihasaGoud (192472346)** (Name / Register Number) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **NIHASGOUD**

Question and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

Definition of Big Data:

Big Data refers to massive, complex datasets whose volume, generation speed (velocity), structural diversity (variety), and uncertainty (veracity) surpass the ingestion, storage, querying, and processing capabilities of conventional relational database management systems (RDBMS).

Why Traditional Data Processing Approaches Fail:

- **Storage and Monolithic Bottlenecks:** Traditional RDBMS systems rely on vertical scaling (Scale-Up), requiring expensive hardware upgrades on single-node servers that eventually hit physical CPU, memory bus, and storage controller ceilings.
- **Schema Rigidity:** Relational databases enforce strict pre-defined tabular schemas (ACID properties), which degrade when attempting to ingest polymorphic, unstructured, or rapidly evolving formats such as PDF resumes, video streams, or semi-structured clickstream logs.
- **Query Latency and Concurrency Limits:** As records scale into millions or billions of rows, table joins, multi-index searches, and aggregate scans trigger severe I/O disk thrashing and query timeouts.

Capstone Context (SkillHire):

In **SkillHire**, processing thousands of unstructured PDF resumes and millions of job board postings concurrently causes single-server database locks. Distributed storage architectures (e.g., Cloud Object Stores) and parallel processing frameworks (e.g., Apache Spark) become essential to distribute compute across a cluster and parse documents simultaneously.

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

The 5Vs of Big Data:

- 1. **Volume:** The sheer physical scale and magnitude of data generated and stored (terabytes to exabytes).
- 2. **Velocity:** The speed at which incoming data is produced, ingested, streamed, and processed in real time or near-real time.
- 3. **Variety:** The structural heterogeneity of incoming data sources (structured tables, semi-structured JSON/XML, unstructured text/video/audio).
- 4. **Veracity:** The degree of noise, data bias, incompleteness, inconsistency, and trustworthiness in raw data feeds.
- 5. **Value:** The actionable business insights, predictive power, and organizational impact extracted from processed data.

Big Data "V"	Concrete Practical Scenario	Impact on Architectural Decision
Velocity	High-Frequency Job Postings & Live User Ingestion: Handling continuous live streams of applicant interactions and newly posted job openings from multiple career platforms during recruitment drives.	Requires distributed message queues (e.g., Apache Kafka / RabbitMQ) and streaming compute engines to ingest and process records without data loss.
Value	Skill Gap Intelligence & Hiring Recommendation: Analyzing thousands of candidate profiles against market demands to recommend targeted upskilling courses and rank qualified candidates for recruiters.	Transforms raw text into measurable career insights, driving faster placement cycles and reducing recruitment costs.

Question 3 (3 Marks)

Differentiate structured, semi-structured, and unstructured data with examples.

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.
3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

Data Type PDF	Architectural Characteristics PDF	Storage / Management Technologies PDF	Concrete Examples (SkillHire / Cloud Context) PDF
Structured Data	Conforms to a fixed tabular schema with explicit column types, constraints, and relational integrity.	Relational Databases (PostgreSQL, MySQL, Cloud SQL).	Recruiter user accounts, transaction logs, applicant registration tables (student_id, email, role).
Semi-Structured Data	Contains self-describing key-value tags, hierarchical tags, and flexible schemas without tabular rigidity.	NoSQL Document Stores (MongoDB, DynamoDB), JSON / XML API payloads.	REST API parsing responses, tokenized skill tags in JSON format ({"skills": ["Python", "FastAPI"], "matchScore": 92}).
Unstructured Data	Has no formal structural model or internal data schema; consists of raw binary or free-form text blobs.	Cloud Object Storage (AWS S3, Firebase Storage), HDFS.	Uploaded candidate PDF/DOCX resumes, candidate portfolio project demo videos, raw scan images.

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.
3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

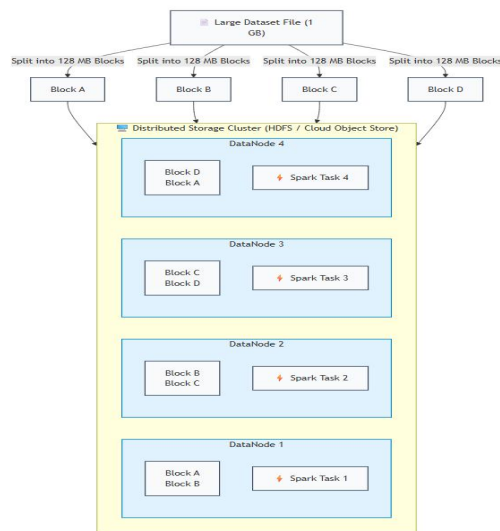
Student Response

Core Role & Architectural Need:

When data scales beyond the capacity of a single physical disk array, a **Distributed File System (DFS)** (e.g., Hadoop Distributed File System — HDFS, or Cloud Object Stores) partitions large datasets into discrete blocks (typically 128 MB or 256 MB) and distributes them across a cluster of commodity worker nodes.

Key Operational Capabilities:

- **Horizontal Scalability:** Storage capacity scales linearly by attaching additional standard storage nodes to the cluster without stopping operations.
- **High Fault Tolerance:** Implements automated data replication (default 3x factor), ensuring continuous availability if individual hardware nodes fail.
- **Data Locality for Parallel Compute:** Enables distributed processing engines (Apache Spark, MapReduce) to execute compute tasks directly on the physical node where the specific data block resides, minimizing high-latency network transfers.



Question 5 (3 Marks)

List any three major security concerns in big data platforms and briefly explain each.

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

Unauthorized Access & Weak Identity Federation:

- *Risk:* Big data lakes consolidate multi-tenant datasets into centralized storage. Without granular access controls, unauthorized users or compromised service accounts can access sensitive organizational data.
- *Mitigation:* Implement strict **Role-Based Access Control (RBAC)**, attribute-based permissions, and mandatory multi-factor authentication (MFA) via OAuth2/JWT tokens.

Data Leakage & Inadequate Encryption:

- *Risk:* Distributing data blocks across multiple cluster nodes increases exposure to packet interception in transit and physical disk extraction at rest.
- *Mitigation:* Enforce end-to-end encryption using **AES-256 at rest** for all block/object stores and **TLS 1.3 in transit** across inter-node cluster channels.

Privacy Violations & Exposure of PII (Personally Identifiable Information):

- *Risk:* Ingesting unstructured documents (e.g., resumes containing phone numbers, physical addresses, candidate names) without sanitization creates regulatory non-compliance risks under GDPR and DPDP laws.
- *Mitigation:* Integrate automated **Data Anonymization, Pseudonymization, and Dynamic PII Masking** pipelines before files enter analytics pipelines.

Question 6 (3 Marks)

What are data appliances and integration tools? Explain their role in big data workflows.

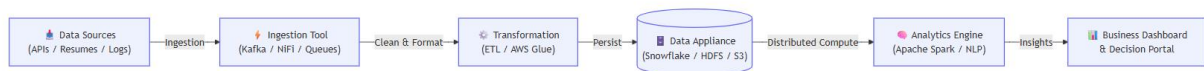
Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Student Response

Data Appliances: Pre-configured, tightly integrated hardware and software appliances optimized for high-throughput data warehousing, indexing, and parallel analytics (e.g., Snowflake, Oracle Exadata, Teradata).

Integration Tools: Middleware platforms and frameworks that automate the ingestion, transformation, synchronization, and movement of data across heterogeneous environments (e.g., Apache NiFi, Airflow, Kafka, AWS Glue).



Question 7 (3 Marks)

Discuss two major challenges organizations face while implementing big data analytics.

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

1. Data Quality and Schema Inconsistency (Veracity Challenge)

- **Cause & Impact:** Raw data collected from disparate external sources (e.g., varied resume formats, duplicate job listings, unstandardized job titles) contains syntax noise, missing values, and conflicting attributes. Training analytical models on poor-quality input leads to inaccurate recommendations and degraded business decisions.
- **Practical Mitigation:** Deploy automated data validation frameworks (e.g., Great Expectations) and schema validation layers during data ingestion to filter and clean records prior to persistence.

2. Infrastructure Costs and Operational Complexity

- **Cause & Impact:** Deploying and maintaining multi-node distributed clusters (storage, memory, GPU compute for NLP/AI) involves high ongoing infrastructure costs and demands specialized skills (DataOps/MLOps). Over-provisioned static clusters lead to unnecessary cloud expenses during low-traffic periods.
- **Practical Mitigation:** Adopt managed, serverless, cloud-native big data architectures (e.g., AWS Athena, Google BigQuery, Databricks) featuring automated elasticity that scales compute down to zero during idle periods.

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.
3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

Selected Domain: Intelligent Human Resources & Talent Analytics (SkillHire Context)

- **Heterogeneous Data Sources:**

Ingests applicant PDF resumes, GitHub repositories, LinkedIn profiles, historical company hiring outcomes, and real-time industry job vacancy descriptions.

- **Analytics and Conversion Pipeline:**

- **Text Mining & NER:** Natural Language Processing (NLP) pipelines parse unstructured text to extract technical proficiencies, certifications, and years of experience.
- **Vector Embeddings & Cosine Matching:** High-dimensional semantic models convert extracted skills into vectors, computing similarity indexes against open role requirements.
- **Predictive Skill Gap Intelligence:** Predictive algorithms identify missing proficiencies within candidate profiles and suggest targeted upskilling tracks.

- **Measurable Organizational Outcomes:**

- Reduces candidate screening turnaround time by over 80%.
- Eliminates recruiter screening bias through objective competency-based ranking.
- Enhances candidate placement rates by aligning job seekers with relevant roles.

Question 9 (3 Marks)

How do business drivers influence the adoption of big data solutions?

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

Business drivers provide the measurable financial, operational, and strategic requirements that justify investing in big data platforms:		
Core Business Driver PDF	Business Objective PDF	Corresponding Technical Big Data Requirement PDF
Operational Efficiency & Cost Reduction	Automate manual document screening workflows and reduce manual labor overhead.	Batch processing pipelines and parallel NLP text-extraction pipelines capable of processing thousands of files per minute.
Hyper-Personalization	Deliver real-time, tailored job and course recommendations to candidates based on profile vectors.	Low-latency vector indexing (FAISS / Pinecone) and machine learning recommendation engines.
Risk Management & Compliance	Protect applicant PII and avoid regulatory penalties under data privacy regulations.	Automated data governance, dynamic PII masking, centralized audit trails, and strict access logging.
Synthesis: Big data implementations should not be driven solely by technology choices; they must be aligned with measurable business outcomes (e.g., faster time-to-hire, reduced churn, lower operating costs).		

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

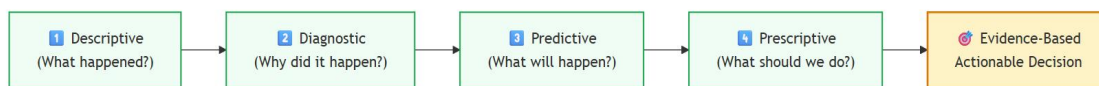
Student Response

Transforming Intuition into Evidence-Based Strategy:

Big data analytics replaces subjective, intuitive decision-making with quantifiable, empirical insights extracted from large data volumes. It enables organizations to make faster, scalable, and automated operational decisions.

The Four Tiers of Analytics-Driven Decision Support:

1. **Descriptive Analytics (What happened?):** Aggregates historical metrics (e.g., *"5,000 candidates applied across 20 open technical roles last month"*).
2. **Diagnostic Analytics (Why did it happen?):** Pinpoints root causes (e.g., *"Recruitment cycle delays were caused by manual PDF text extraction bottlenecks"*).
3. **Predictive Analytics (What is likely to happen?):** Employs statistical forecasting models (e.g., *"Demand for Cloud Security Engineers will grow by 45% next quarter"*).
4. **Prescriptive Analytics (What action should be taken?):** Automated decision recommendations (e.g., *"Reallocate recruitment budget to cloud hiring channels and suggest Docker/Kubernetes training tracks to applicants"*).



Submission Checklist

Checklist Item	Status	Remarks
Student details completed.	Done	Nothing
All 10 questions answered.	Done	Nothing
Examples are connected to cloud/capstone/domain context.	Done	Nothing
Security/privacy points are included wherever relevant.	Done	Nothing
Document exported as PDF if requested.	Done	Nothing
GitHub link shared through Google Classroom.	Done	Nothing